

CHAPTER II: LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of literature related to criminal case management systems, digital transformation in law enforcement institutions, web-based information systems, blockchain technology, and secure digital evidence management. The purpose of this chapter is to examine previous scholarly studies, technological developments, and practical implementations relevant to the proposed Police Station Management System (PSMS).

The rapid advancement of Information and Communication Technology (ICT) has transformed the operational structures of organizations globally, including institutions within the criminal justice sector. Law enforcement agencies increasingly depend on digital technologies to improve operational efficiency, transparency, accountability, communication, and data security. Traditional paper-based methods are gradually being replaced by computerized systems capable of managing large volumes of sensitive criminal information effectively.

Criminal case management systems play an important role in supporting police investigations, evidence handling, suspect tracking, and administrative coordination. Efficient criminal information management contributes significantly to justice administration, public trust, institutional accountability, and crime prevention. Poor management of criminal records may lead to delayed investigations, loss of evidence, information inconsistency, corruption, and weak coordination among security institutions.

Modern policing environments require integrated digital platforms capable of supporting real-time communication, centralized data management, and secure information sharing across administrative levels. As a result, many governments and security agencies have adopted web-based management systems to automate criminal case registration, officer deployment, evidence tracking, and reporting activities.

Recently, blockchain technology has emerged as one of the most promising innovations for improving digital security and data integrity. Blockchain provides decentralized and tamper-resistant data storage mechanisms that help protect sensitive information from unauthorized modifications. In criminal justice environments, blockchain technology supports evidence protection, audit logging, chain of custody management, identity verification, and accountability mechanisms.

Despite these technological advancements, many developing countries still face major challenges in implementing secure and integrated criminal case management systems. In Somalia, numerous police stations continue to rely on manual or semi-digital processes characterized by fragmented

records, weak security mechanisms, limited accessibility, and poor coordination among administrative institutions. These limitations create opportunities for information loss, unauthorized data manipulation, operational inefficiency, and corruption risks.

The proposed Police Station Management System is designed to address these challenges by integrating modern web technologies with blockchain-based integrity mechanisms. The system introduces a structured five-tier administrative hierarchy that controls access to information and ensures secure management of criminal investigations across multiple organizational levels.

This chapter therefore reviews relevant literature related to:

- Criminal Case Management Systems (CCMS)
- Manual record-keeping systems in policing
- Digital transformation in law enforcement
- Web-based information systems
- Blockchain technology
- Digital evidence and chain of custody management
- Authentication and cybersecurity mechanisms
- Hierarchical information systems
- Related studies and existing research gaps

The chapter concludes by identifying limitations in existing systems and justifying the need for a secure, scalable, blockchain-integrated criminal case management system tailored for Somali law enforcement institutions.

2.2 Overview of Criminal Case Management Systems

A Criminal Case Management System (CCMS) refers to a computerized platform designed to support the recording, tracking, monitoring, investigation, and reporting of criminal cases throughout their lifecycle. These systems are widely used by police departments, investigative agencies, courts, correctional institutions, and prosecutors to improve criminal justice operations.

The primary objective of a CCMS is to improve operational efficiency while ensuring secure and accurate documentation of criminal investigations. Modern systems automate administrative activities and reduce reliance on manual paperwork.

A standard criminal case management system commonly includes the following functionalities:

- Criminal case registration

- Occurrence Book (OB) generation
- Suspect management
- Victim profiling
- Witness statement recording
- Evidence documentation
- Investigation tracking
- Officer assignment
- Reporting and analytics
- Access control
- Audit logging
- Case lifecycle monitoring

Modern CCMS platforms typically rely on centralized databases and web-based interfaces that allow authorized personnel to retrieve and update information in real time. These systems support collaboration among departments and improve institutional coordination.

Research indicates that digital criminal management systems significantly reduce record duplication, improve accessibility, minimize human errors, and strengthen accountability mechanisms. Electronic systems also improve crime analysis and strategic decision-making through automated reporting and statistical analysis.

However, the effectiveness of a CCMS depends on several factors including:

- System security
- Infrastructure reliability
- User training
- Organizational readiness
- Data integrity protection
- Scalability

Poorly designed systems remain vulnerable to unauthorized access, cyberattacks, and data tampering. Therefore, modern criminal management systems increasingly integrate cybersecurity technologies and cryptographic protection mechanisms.

2.3 Traditional Manual Record-Keeping Systems in Law Enforcement

Before the emergence of computerized systems, police institutions relied heavily on paper-based documentation to manage criminal investigations and administrative records. Police officers manually recorded incidents in physical registers, files, and archives.

Although manual systems were initially manageable in small-scale environments, they became increasingly inefficient as criminal records expanded.

2.3.1 Risk of Data Loss

Paper-based records are vulnerable to:

- Fire damage
- Flooding
- Theft
- Physical deterioration
- Human negligence

Missing records can delay investigations and weaken institutional accountability.

2.3.2 Slow Information Retrieval

Manual archives require significant time and human effort to retrieve information. Officers often experience delays when searching historical case records or generating reports.

2.3.3 Data Duplication and Inconsistency

Repeated manual data entry frequently creates duplicate files and inconsistent information across departments.

2.3.4 Weak Security Mechanisms

Traditional systems often lack proper access control mechanisms. Unauthorized individuals may access confidential files, and documents can be altered without detection.

2.3.5 Limited Coordination

Manual systems make information sharing between police departments difficult, reducing operational collaboration and investigation effectiveness.

These challenges have encouraged governments worldwide to adopt digital transformation strategies aimed at modernizing law enforcement operations.

2.4 Digital Transformation in Law Enforcement

Digital transformation refers to the integration of digital technologies into organizational processes to improve operational efficiency, service delivery, communication, and decision-making.

In policing environments, digital transformation involves replacing manual procedures with electronic systems capable of supporting modern law enforcement activities.

Over the past two decades, law enforcement institutions have adopted technologies such as:

- Web-based management systems
- Cloud computing
- Geographic Information Systems (GIS)
- Artificial intelligence
- Digital forensics
- Blockchain technology
- Biometric authentication systems

2.4.1 Improved Operational Efficiency

Digital systems automate repetitive tasks and reduce paperwork, improving workflow management.

2.4.2 Faster Communication

Integrated platforms support real-time communication between officers, departments, prosecutors, and courts.

2.4.3 Enhanced Transparency and Accountability

Electronic systems maintain traceable records of all transactions and activities, reducing corruption risks.

2.4.4 Better Data Analytics

Modern platforms generate crime statistics and reports that support evidence-based policing and strategic planning.

2.4.5 Increased Accessibility

Authorized users can access centralized databases remotely regardless of location.

Despite these advantages, many developing countries continue to face challenges including:

- Limited ICT infrastructure
- Financial constraints
- Weak cybersecurity frameworks
- Lack of technical expertise
- Organizational resistance to change

2.5 Hierarchical Information Systems in Law Enforcement

Hierarchical information systems are organizational systems where access permissions and operational visibility are structured according to administrative levels.

The proposed Police Station Management System adopts a strict five-tier administrative hierarchy consisting of:

1. State Administration
2. Regional Administration
3. City Administration
4. District Administration
5. Neighborhood/Police Station

Hierarchical systems provide several advantages:

2.5.1 Controlled Data Visibility

Users can only access information within their administrative scope and subordinate levels.

2.5.2 Improved Security

Sensitive criminal information remains compartmentalized, reducing unauthorized exposure.

2.5.3 Better Officer Management

Commanders can supervise officers assigned within their administrative jurisdictions.

2.5.4 Efficient Organizational Coordination

Hierarchical systems support structured communication and reporting channels.

2.5.5 Accountability and Oversight

Administrative supervisors can monitor activities occurring within their jurisdiction.

Strict hierarchical separation is particularly important in criminal justice systems because it prevents unauthorized access to investigations and protects operational confidentiality.

2.6 Web-Based Information Systems

A web-based information system is a software application accessed through internet browsers and hosted on servers.

Web technologies have become highly popular because they provide:

- Scalability
- Remote accessibility
- Multi-user support
- Centralized updates
- Cross-platform compatibility

Web-based criminal management systems allow officers to:

- Register criminal cases online
- Upload evidence
- Track investigations
- Generate reports
- Monitor officer assignments

- Coordinate interdepartmental activities

The architecture of web-based systems generally consists of:

1. Frontend Interface
2. Backend Application Logic
3. Database Management System
4. Network Infrastructure

The proposed Police Station Management System uses:

- Next.js and React for frontend development
- Node.js and Express.js for backend services
- MySQL relational database
- RESTful APIs for communication

Despite their advantages, web-based systems face several cybersecurity threats including:

- SQL injection attacks
- Unauthorized access
- Malware attacks
- Data breaches
- Session hijacking
- Data tampering

Therefore, integrating advanced security mechanisms is essential.

2.7 Authentication and Access Control Systems

Authentication and access control mechanisms are critical components of secure criminal management systems.

Authentication refers to verifying user identity before granting access to system resources.

Modern systems commonly use:

- Username/password authentication
- Multi-factor authentication
- JSON Web Tokens (JWT)

- Biometric verification

The proposed system uses:

- JWT-based stateless authentication
- bcrypt password hashing
- Role-based access control (RBAC)

Role-Based Access Control ensures users only access information relevant to their responsibilities.

System roles include:

- System Administrator
- State Commander
- Regional Commander
- District Commander
- CID Investigator
- Prosecutor
- Jail Officer
- Station Officer

Access control significantly improves:

- Security
- Accountability
- Operational integrity
- Confidentiality

2.8 Blockchain Technology

Blockchain is a distributed ledger technology that stores records in linked cryptographic blocks.

Each block contains:

- Transaction data
- Timestamp
- Cryptographic hash
- Previous block hash

Blockchain possesses several important characteristics.

2.8.1 Decentralization

Data is distributed across multiple nodes instead of a single centralized server.

2.8.2 Immutability

Recorded data cannot easily be altered without consensus.

2.8.3 Transparency

Transactions can be verified and audited.

2.8.4 Security

Cryptographic hashing protects records from tampering.

2.8.5 Traceability

Blockchain maintains permanent historical records.

These characteristics make blockchain highly suitable for criminal justice systems where data integrity and accountability are essential.

2.9 Blockchain Applications in Criminal Justice Systems

Researchers increasingly explore blockchain applications in law enforcement and judicial environments.

2.9.1 Evidence Management

Blockchain secures digital evidence and maintains chain of custody records.

2.9.2 Criminal Record Protection

Sensitive records can be protected against unauthorized modifications.

2.9.3 Audit Logging

Blockchain provides transparent logs for user activities and system transactions.

2.9.4 Identity Verification

Blockchain-based authentication improves trust and verification mechanisms.

2.9.5 Smart Contracts

Smart contracts automate permissions and access control operations.

The proposed Police Station Management System integrates blockchain integrity verification using SHA-256 cryptographic hashing for:

- Case creation
- Case transfer
- Evidence updates
- Confirmation activities

This guarantees that critical criminal records cannot be secretly modified.

2.10 Evidence Management and Chain of Custody

Chain of custody refers to the chronological documentation of evidence handling from collection to court presentation.

Proper chain of custody management ensures:

- Evidence authenticity
- Accountability
- Court admissibility
- Investigation transparency

Digital evidence management systems help track:

- Evidence ownership
- Transfer history

- Access logs
- Storage locations

Blockchain integration strengthens chain of custody management by creating immutable transaction histories.

2.11 Audit Logging Systems

Audit logging refers to recording all user activities and system transactions.

Examples include:

- User logins
- Case updates
- Evidence uploads
- Officer transfers
- Record deletions

Audit logs improve:

- Transparency
- Accountability
- Security monitoring
- Incident investigation

The proposed system records:

- User ID
- Timestamp
- Action description
- IP information
- Administrative location

2.12 Officer and Personnel Management Systems

Personnel management systems help institutions manage:

- Officer records

- Ranks
- Assignments
- Transfers
- Deployment history

The proposed system includes centralized officer management supporting:

- Dynamic deployments
- Transfer tracking
- Administrative visibility filtering
- Command hierarchy supervision

These capabilities improve workforce coordination and accountability.

2.13 Related Work

Several studies have explored blockchain-based criminal management systems and digital evidence platforms.

Alyas et al. (2025) proposed a multi-blockchain judicial management architecture using smart contracts to improve scalability and transparency.

Liu and Zheng (2024) developed a blockchain-based judicial evidence preservation framework integrating off-chain storage for secure evidence handling.

Ratul et al. (2024) examined blockchain-based digital evidence management in cybercrime investigations using Ethereum and Polygon technologies.

Patil et al. (2024) explored blockchain applications for maintaining forensic chain of custody.

Previous studies on web-based police systems demonstrate improvements in:

- Operational efficiency
- Record accessibility
- Automated reporting
- Administrative coordination

However, many existing systems either:

- Lack strong tamper-proof security mechanisms, or
- Focus on blockchain without practical law enforcement workflow integration.

Very limited research specifically addresses Somalia’s policing environment.

2.14 Comparison of Existing Studies

Table 2.1 Comparison of Existing Studies

Author & Year	System Focus	Technology Used	Strengths	Limitations
Alyas et al. (2025)	Judicial Case Management	Multi-Blockchain & Smart Contracts	Transparency and scalability	Limited police station focus
Liu & Zheng (2024)	Evidence Preservation	Blockchain + Off-chain Storage	Strong immutability	Does not manage full case lifecycle
Ratul et al. (2024)	Cybercrime Evidence Management	Ethereum/Polygon Blockchain	Tamper-proof evidence handling	Focused on cybercrime only
Patil et al. (2024)	Forensic Chain of Custody	Blockchain Ledger Technology	Evidence traceability	Limited practical implementation

2.15 Challenges Facing Criminal Case Management Systems in Somalia

Somali law enforcement institutions face several technological and operational challenges.

2.15.1 Dependence on Manual Systems

Many police stations still rely on paper-based documentation.

2.15.2 Weak Information Security

Sensitive records remain vulnerable to unauthorized access and manipulation.

2.15.3 Lack of Centralized Databases

Information sharing remains fragmented.

2.15.4 Limited Infrastructure

Insufficient ICT infrastructure limits digital transformation.

2.15.5 Inadequate Technical Training

Many officers lack digital system management skills.

2.15.6 Poor Coordination

Interdepartmental communication is often inefficient.

These challenges highlight the urgent need for a secure and scalable digital solution.

2.16 Research Gap

The reviewed literature demonstrates that digital criminal management systems improve operational efficiency, while blockchain technology enhances security, transparency, and integrity.

However, several important gaps remain.

First, many web-based criminal systems lack strong tamper-proof mechanisms capable of protecting sensitive records from unauthorized modifications.

Second, blockchain-focused studies often remain conceptual without practical integration into daily police workflows.

Third, existing systems rarely implement strict hierarchical visibility structures suitable for law enforcement institutions.

Fourth, limited research addresses operational realities within developing countries, particularly Somalia.

Finally, few systems integrate:

- Criminal case tracking
- Blockchain integrity verification
- Officer deployment management
- Audit logging

- Chain of custody management
- Hierarchical access control

within a single scalable platform.

Therefore, this study aims to fill these gaps by developing a secure, scalable, blockchain-integrated Police Station Management System specifically tailored to Somali law enforcement institutions.

The proposed system seeks to improve:

- Criminal investigation management
- Evidence integrity
- Transparency and accountability
- Institutional coordination
- Officer supervision
- Secure information sharing

References

Alyas, T., Abbas, Q., Niazi, S., Alqahtany, S. S., Alghamdi, T., Alzahrani, A., Tabassum, N., & Ibrahim, A. M. (2025). Multi blockchain architecture for judicial case management using smart contracts. *Scientific Reports*, 15(1).

Liu, S., & Zheng, Q. (2024). A study of a blockchain-based judicial evidence preservation scheme. *Blockchain: Research and Applications*, 5(2).

Patil, H., Kohli, R. K., Puri, S., & Puri, P. (2024). Potential applicability of blockchain technology in the maintenance of chain of custody in forensic casework. *Egyptian Journal of Forensic Sciences*, 14(1).

Ratul, M. H. A., Mollajafari, S., & Wynn, M. (2024). Managing Digital Evidence in Cybercrime: Efforts Towards a Sustainable Blockchain-Based Solution. *Sustainability*, 16(24).

Bharadwaj, A. (2000). A resource-based perspective on information technology capability and firm performance.

DeLone, W. H., & McLean, E. R. (2003). The DeLone and McLean model of information systems success.

Laudon, K. C., & Laudon, J. P. (2014). *Management Information Systems*.

Pressman, R. S. (2010). *Software Engineering: A Practitioner's Approach*.

Sommerville, I. (2011). *Software Engineering*.

Turban, E., Volonino, L., & Wood, G. (2011). *Information Technology for Management*.